

Independent Evaluation of RETFound Foundation Model's Performance on Optic Nerve Analysis Using Fundus Photography

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Purpose: This study evaluates RETFound, a retinal image foundation model, as a feature extractor for predicting optic nerve metrics like cup-to-disc ratio (CDR) and retinal nerve fiber layer (RNFL) thickness using an independent clinical dataset.

Design: Retrospective observational study.

Participants: Patients who underwent fundus photography and RNFL OCT at the Byers Eye Institute, Stanford University.

Methods: Fundus images were paired with RNFL OCT results where study dates were within 6 months of each other. Latent features from full-sized raw fundus images were extracted from RETFound and used as inputs for several linear regression models (Ridge, Lasso, Elastic Net, and ordinary least squares). Baseline models using pretrained VGG16 and Vision Transformers (ViTs) as feature extractors were also developed. All models were trained to perform single-output tasks (predicting CDR or average RNFL thickness) and multioutput tasks (predicting RNFL thickness at quadrants and clock hours). Data were split 80:20 at the patient level for training and validation.

Main Outcome Measures: Model predictions were evaluated on a test set using the metrics of R^2 , mean absolute error, and root mean square error.

Results: Among the 463 unique participants, contributing 776 fundus–OCT data pairs, the mean age was 63 years (± 18 years), with 57.24% being female ($N = 265$). RETFound models demonstrated strong performance on single-output tasks, achieving R^2 values between 0.706 and 0.898 for CDR prediction and between 0.855 and 0.961 for average RNFL thickness prediction. Performance on multioutput tasks was less robust, with a highest R^2 of 0.583 for clock-hour RNFL thickness prediction and an R^2 of 0.811 for quadrant RNFL thickness prediction. RETFound models outperformed VGG16 and ViT models, which achieved maximum R^2 of 0.731 and 0.687 in predicting RNFL thickness and CDR.

Conclusions: Machine learning models leveraging the massively pretrained RETFound foundation model could accurately predict CDR and average RNFL thickness from fundus photos on an independent clinical dataset. Although RETFound was not trained or fine-tuned for these optic nerve evaluation tasks, nevertheless, RETFound overcomes small dataset limitations and excels in specialized applications.

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Supplemental material available at www.ophtalmologyscience.org.

Glaucoma is a leading cause of irreversible blindness globally.¹ Many patients with glaucoma are asymptomatic in the early stages, which often leads to diagnosis at late stages of the disease, by which time irreparable damage to the optic nerve has already occurred.² Therefore, strategies that aid in the early diagnosis of glaucoma are needed. Currently, glaucoma is diagnosed with the assistance of structural and functional assessments of the optic nerve. Structural changes of the optic nerve are key signs of glaucomatous damage and include thinning of the nerve fiber layer and posterior bowing of the lamina cribrosa.^{1,3,4} OCT is an imaging method that provides objective, repeatable measurements of the optic nerve at a

high resolution.^{5,6} These include morphologic measurements of the optic nerve as well as measurements of the thickness of the nerve fiber layer. OCT imaging of the nerve is a specialized imaging modality that is relatively high cost and not universally available in all settings. Strategies that can extract these key optic nerve features from more accessible fundus imaging, such as color fundus photographs of the optic nerve, would be useful for glaucoma screening.

Several techniques leveraging artificial intelligence for early diagnosis of glaucoma from fundus photography are currently in development. These include convolutional neural networks,⁷ transformers,⁸ and other deep learning

architecture.^{9,10} Most of these models are trained to accomplish a single task, such as identifying cup-to-disc ratio (CDR)^{11–15} or retinal nerve fiber layer (RNFL) thickness.^{14–17} These methods have required large training datasets with clinician-generated labels, which require significant amounts of time to prepare and computing costs to train. To address these limitations, self-supervised learning offers a powerful approach by utilizing vast amounts of unlabeled data to pretrain models.^{18,19} This initial learning phase enables the model to capture rich feature representations. Afterward, the model can be fine-tuned on smaller, labeled datasets for specific downstream tasks, or the pretrained self-supervised model can be employed as a feature extractor, enhancing performance on tasks where labeled data is scarce.^{18,19} This approach not only maximizes the utility of available data but also improves the model's generalization and adaptability across different applications.

RETFound, a recently released foundation model for retinal images, has utilized self-supervised learning by training on 1.6 million unlabeled OCT images and fundus photographs from Moorfields Eye Hospital and University College London.²⁰ The potential impact of RETFound lies in its ability to be fine-tuned or be used as a feature extractor for a wide variety of downstream regression or classification tasks such as RNFL thickness, diabetic retinopathy, and glaucoma.²⁰ Although RETFound's performance has been validated on several existing public datasets on a variety of disease evaluation tasks, there remains a need to assess its performance on independent clinical datasets and on diverse downstream tasks to evaluate its generalizability. The purpose of this study was to leverage the pretrained RETFound model to extract features from fundus photography to perform a variety of optic nerve evaluation tasks, thus evaluating RETFound on an independent clinical dataset and on tasks that it was not originally trained to perform. Using RETFound, we developed models to identify CDR and RNFL thickness (overall average, quadrants, and clock hours) from fundus photographic images. By using RETFound to extract key features from fundus photographs, a limited dataset can be used to specifically train the model for these tasks, reducing the imaging corpus needs while maintaining strong predictive performance.

Methods

Data Source

We identified standard 45-degree macula-centered color fundus photos for patients seen at the Stanford Byers Eye Institute between March 2014 and October 2021. Photos where the optic nerve was not completely in the field of view or where the image was obviously out of focus were excluded. OCT images with signal strength <6 were excluded. Images with signal strength below 6 are often blurry or have artifacts that can affect analysis.²¹ Key retinal features may not be clearly visible in these low-quality images.²¹ Removing them ensures the model learns from clear accurate examples. Setting a minimum signal strength of 6 maintains consistent quality across datasets, improving result comparisons.²¹

We included only fundus images with corresponding RNFL OCT studies for that eye within 6 months. In cases where there were multiple RNFL OCT studies for the photographed eye within 6 months, we matched the photo with the OCT study that was nearest in time. Fundus images were captured by Carl Zeiss Meditec FF450 plus machines, and RNFL OCT scans were captured by Zeiss Cirrus 5000 machines. We began with 1351 fundus images and 18 233 RNFL reports. After matching the fundus images with the corresponding RNFL reports, the final cohort comprised 776 photo–RNFL OCT pairs belonging to 463 unique patients. Fundus images were resized to 224×224 and normalized with ImageNet mean and standard deviation (SD). Bicubic interpolation and center crop were also performed on the images. These samples were split 80:20 at the patient level for training and validation. Figure 1 shows an overview of dataset creation. This study adheres to the tenets of the Declaration of Helsinki and was approved by the Stanford Institutional Review Board with a waiver of informed consent.

Modeling

We developed machine learning prediction models to evaluate the ability of RETFound as a feature extractor to perform 4 tasks on color fundus photos: (1) identifying CDR, (2) predicting average RNFL thickness, (3) predicting quadrant RNFL thickness (temporal, nasal, superior, and inferior), and (4) predicting clock-hour RNFL thickness (over each of 12 clock hours). Results from the RNFL OCT scans were used as the ground truth for training purposes. Clock hours were normalized to right eye configuration, such that 3 o'clock always represents temporal thickness and 9 o'clock represents nasal thickness. We employed the pretrained RETFound as a feature extractor for fundus images. The images were processed through RETFound, with the RETFound model weights frozen, ensuring that the feature extraction remained consistent and unbiased by the subsequent regression models. The extracted latent features, which is a vector of 128 features, were then used as inputs for a variety of linear regression models trained to perform the above 4 tasks. Specifically, we explored the performance of (1) ordinary least squares, which minimizes the sum of squared residuals to find the best-fitting line; (2) ridge regression, which adds an L2 penalty to the loss function to prevent overfitting; (3) lasso regression, which incorporates an L1 penalty to promote sparsity in the feature set; and (4) Elastic Net, a

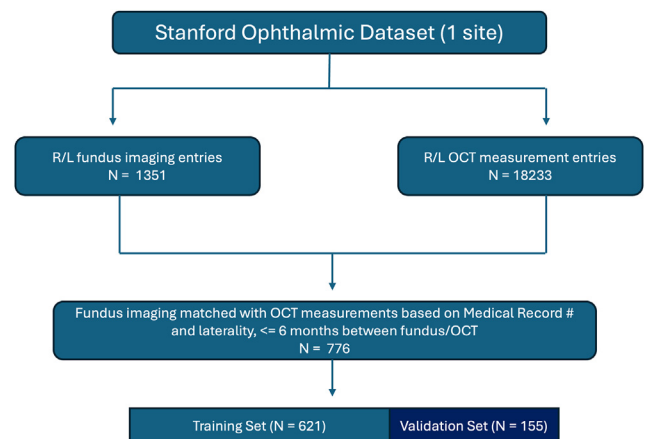


Figure 1. Cohort creation flow diagram. This figure gives an overview of cohort creation and identification of images for model training. R/L = right/left.

combination of both L1 and L2 penalties that balances between ridge and lasso regression. This approach allowed us to compare the effectiveness of different regression techniques in predicting the desired outcomes based on the features extracted by RETFound. Depending on the task at hand, the linear regression models were single output or multioutput; for example, for predicting CDR or average RNFL thickness, the model produces a single prediction, whereas, for predicting RNFL thickness across the quadrants or the 12 clock hours, the model produces a vector containing multiple predictions corresponding to each location. Hyperparameters were tuned using fivefold cross-validation on the training set to minimize mean squared error (MSE). Figure 2 illustrates the overall training and inference pipeline.

We also evaluated 2 additional feature extractors for baseline comparison: a pretrained Vision Transformer (ViT)²² (vit-base-patch16-224) model and a pretrained VGG16²³ convolutional network. The “vit-base-patch16-224” refers to a ViT model with a base architecture that splits images into 16×16 patches and processes them with an input resolution of 224×224 pixels. This allows the model to analyze smaller image regions in detail, making it effective for image classification tasks. All baselines and RETFound follow the same pipelines when being adapted to the 4 prediction tasks. Both models utilized pretrained weights from the ImageNet dataset, which includes 1000 object classes derived from 1.2 million general domain images. Hyperparameters for all models are reported in Table S1 (available at www.opthalmology-science.org). Only hyperparameters that have been explicitly changed from default values are reported.

Evaluation

We evaluated all models using standard regression evaluation metrics including MSE, mean absolute error (MAE), root mean squared error (RMSE), and R^2 on the held-out test set. Mean squared error measures the average squared difference between predicted and actual values, giving higher weight to larger errors, making it sensitive to outliers. Mean absolute error calculates the average absolute difference between predicted and actual values, providing a straightforward interpretation of prediction accuracy in the same units as the target variable. Root mean squared error is the square root of MSE, offering an error measure in the same units as the target variable and making it easier to interpret. R^2 , or the coefficient of determination, indicates the proportion of the variance in the dependent variable that is predictable from the

independent variables, with values closer to 1 indicating better model performance.

Results

Population Characteristics

Patient and eye characteristics are summarized in Table 2. A total of 776 pairs of fundus and RNFL images were included, belonging to 463 unique patients. The mean age of patients was 63 years old with a SD of 18 years. Of the unique patients, 57.24% of the cohort ($N = 265$) were women, 42.33% ($N = 196$) were White, 28.29% ($N = 131$) were Asian, and 12.31% ($N = 57$) were Hispanic/Latino patients. The mean average RNFL thickness was 90.4 (SD: 31.7), and the mean CDR was 0.5 (SD: 0.2).

Single-Output Regression Tasks: Average RNFL Thickness and CDR

Table 3 presents the performance results of regression models for identifying average RNFL thickness from fundus photos on the held-out test set. Modeling approaches incorporating RETFound as a feature extractor demonstrated strong performance, with R^2 values ranging from 0.855 to 0.961, with the RETFound + Ridge variant achieving the highest accuracy. Baseline models performed poorly, including the ViT model which achieved an R^2 of 0.731 and the VGG16 model which achieved an R^2 of 0.600. Figure 3 shows predicted vs. actual RNFL for the baseline models and the best-performing ridge model (Panel A) and the prediction residuals (Panel B). The RETFound + Ridge model’s residuals are clustered more closely around zero, indicating more accurate predictions. In comparison, the residuals for the ViT and VGG16 models are more widely dispersed, reflecting their lower predictive accuracy.

Model performance on the CDR task showed a similar pattern, with RETFound models outperforming baseline

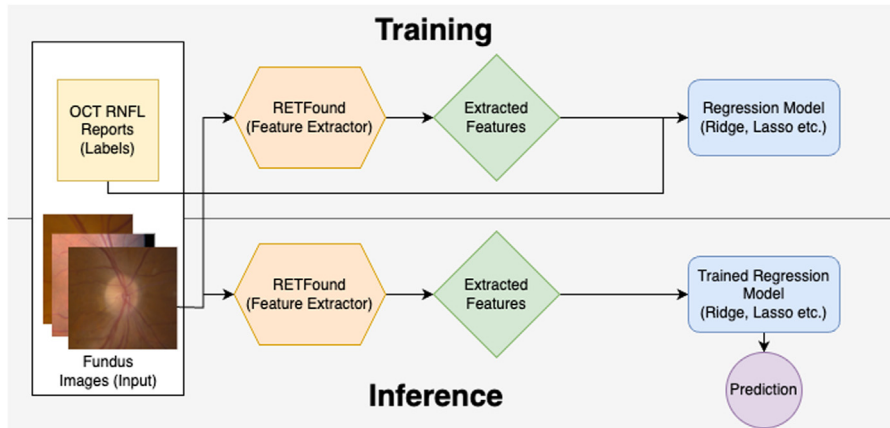


Figure 2. Model training and inference pipelines. This figure shows an overview of the model training and inference pipeline. Fundus images are input into the pretrained RETFound model, and resulting extracted features serve as inputs into machine learning regression models that predict CDR and RNFL thickness, determined from matching OCT RNFL reports. CDR = cup-to-disc ratio; RNFL = retinal nerve fiber layer.

Table 2. Population Characteristics

		Total Sample (N = 776)	
		N	%
Laterality	Left	291	37.5%
	Right	485	62.5%
		Mean	Std
Avg RNFL thickness		90.4	31.7
Avg rim area		1.3	0.4
Avg disc area		1.9	0.5
Avg CDR		0.5	0.2
Vertical CDR		0.5	0.2
Cup volume		0.23	0.2
Temporal		66	22.1
Superior		108	42.3
Nasal		72.8	29.8
Inferior		114.9	47.9
Clock hour 1		97.3	44
Clock hour 2		87.8	37.2
Clock hour 3		63.1	26.4
Clock hour 4		67.6	31
Clock hour 5		95.5	49
Clock hour 6		124.7	56.6
Clock hour 7		124.4	48.6
Clock hour 8		66.8	24.4
Clock hour 9		54.5	20.1
Clock hour 10		76.6	29.5
Clock hour 11		119.1	44.5
Clock hour 12		107.6	50
		Unique participants (N = 463)	
		Mean	Std
Age		63	18
		N	%
Sex			
Male		198	42.76%
Female		265	57.24%
Race			
White		196	42.33%
Black or African American		14	3.02%
Asian		131	28.29%
American Indian or Alaska Native		1	0.22%
Native Hawaiian or other Pacific Islander		3	0.65%
Other		93	20.09%
Unknown		17	3.67%
Ethnicity			
Hispanic/Latino		57	12.31%
Non-Hispanic/non-Latino		387	83.59%
Unknown		11	2.38%

Avg = average; CDR = cup-to-disc ratio; RNFL = retinal nerve fiber layer.

models (Table 3). The RETFound + lasso model achieved the highest R^2 value of 0.898. In comparison, the ViT model obtained an R^2 of 0.687, and the VGG16 model performed worst with an R^2 of 0.499. Figures 4A, B show

Table 3. Average RNFL Thickness and CDR Model Results

Model	MAE	MSE	RMSE	R^2
Average RNFL thickness				
RetFound + ridge	3.876	28.274	5.317	0.961
RetFound + lasso	4.037	29.398	5.422	0.959
RetFound + ElasticNet	4.141	30.784	5.548	0.957
RetFound + OLS	7.799	105.080	10.251	0.855
Pretrained ViT fine-tuned	9.944	194.281	13.938	0.731
Pretrained VGG fine-tuned	10.811	289.018	17.001	0.600
CDR				
RetFound + ridge	0.079	0.010	0.099	0.833
RetFound + lasso	0.060	0.006	0.077	0.898
RetFound + ElasticNet	0.064	0.007	0.085	0.878
RetFound + OLS	0.103	0.017	0.131	0.706
Pretrained ViT fine-tuned	0.111	0.018	0.135	0.687
Pretrained VGG fine-tuned	0.143	0.029	0.171	0.499

CDR = cup-to-disc ratio; MAE = mean absolute error; MSE = mean squared error; RMSE = root mean squared error; RNFL = retinal nerve fiber layer; ViT = Vision Transformer.

Bolded values correspond to the metrics from the strongest-performing regression model.

predicted vs. actual CDR and residuals. The residuals of the RETFound + Lasso model are more tightly clustered around 0 compared with those of the ViT and VGG16 models, indicating superior predictive accuracy.

Multitask Regression Tasks: RNFL Thickness by Quadrants and Clock Hours

For predicting RNFL thickness by clock hour, which involves predicting 12 outputs for each input photo, our models demonstrate varying levels of performance (Table 4). The RETFound + ElasticNet model was the top performer, with an overall R^2 value of 0.583, followed closely by RETFound + lasso ($R^2 = 0.582$) and RETFound + ridge ($R^2 = 0.572$). These models achieve MAE values around 13.2 to 13.3 and RMSE values around 19.6 to 19.8. The ViT and VGG models show poorer performance, with an R^2 value of 0.444 and 0.340, respectively. Table S5 (available at www.opthalmologyscience.org) provides the R^2 for each clock hour, alluding that clock hours 12 and 6 performed the best with RETFound + ElasticNet having an R^2 of 0.912 and 0.936, respectively.

For predicting RNFL thickness by quadrant, which involves predicting 4 outputs for each input photo, RETFound models again outperformed baseline models. The RETFound + lasso model achieves the highest R^2 value of 0.811, closely followed by RETFound + ridge ($R^2 = 0.809$) and RETFound + ElasticNet ($R^2 = 0.808$). These models also display low MAE values around 3.4 and RMSE values around 7.8, indicating high accuracy in their predictions. Our baseline models showed much poorer results with ViT having an R^2 score of 0.732 and VGG16 having an R^2 score of 0.668. Table S6 (available at www.opthalmologyscience.org) provides the R^2 for each quadrant, with superior and inferior having the best predictions. RETFound + lasso had an R^2 of 0.99 and 0.99, respectively.

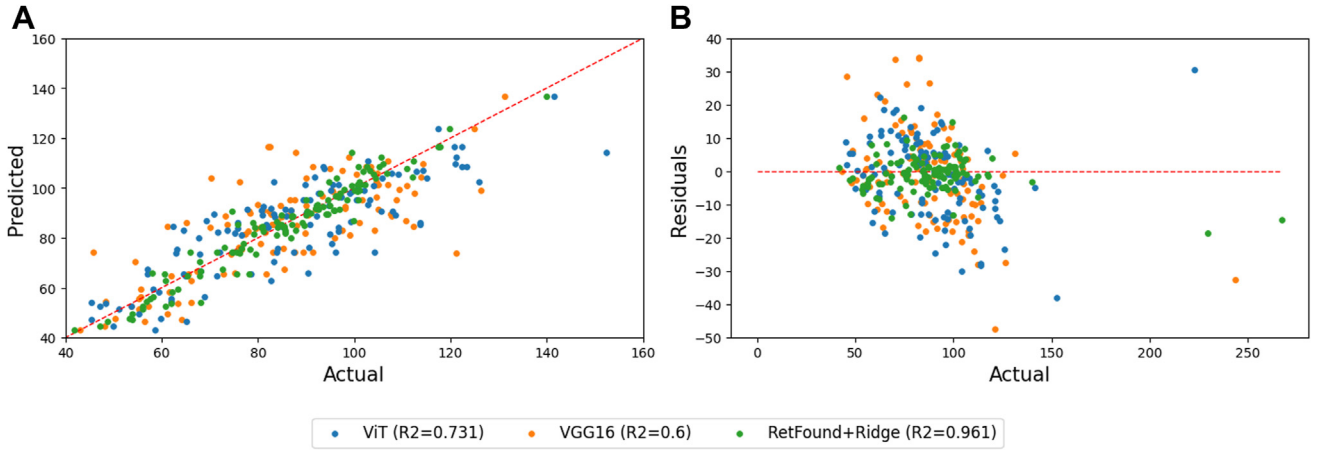


Figure 3. Regression performance for retinal nerve fiber layer (RNFL) thickness. This figure shows scatterplot visualizations of regression results for models predicting RNFL Thickness, including the best RETFound model and 2 baseline models (ViT and VGG16). **A**, The individual model-predicted RNFL thickness values against the actual values. **B**, The individual residuals for each sample. ViT = Vision Transformer.

Performance Stratified by Race/Ethnicity

We also analyzed the performance of each task stratified by race/ethnicity. For all tasks, non-Hispanic White performed the best with R^2 scores of 0.982 (RNFL thickness), 0.918 (CDR), 0.592 (clock hours), and 0.872 (clock quadrants). The non-Hispanic Asian population was the second best in terms of R^2 with scores of 0.933 (RNFL thickness), 0.862 (CDR), 0.562 (clock hours), and 0.801 (clock quadrants). Results are displayed in Table 7.

Discussion

In this study, we aimed to validate the ophthalmic foundation model, RETFound, on an independent dataset and assess its performance on glaucoma-related tasks for which it was not originally trained: predicting optic nerve CDR and

RNFL thickness from fundus images. By using RETFound as a feature extractor and using those extracted features as inputs into regression models, our results demonstrate that RETFound significantly outperforms baseline models across all tasks. This underscores the versatility of foundation models like RETFound for downstream tasks beyond their initial training. Additionally, our analysis revealed that model performance among non-Hispanic White and Asian populations was highest across all tasks.

Using the massively pretrained RETFound as the backbone of machine learning models, we were able to identify CDR and predict average RNFL thickness from fundus photography with performance comparable to or exceeding prior efforts, despite a relatively small corpus of images from an independent site. Most prior models use classical convolutional neural networks, trained on exceptionally large single-center corpuses of images.^{11,24–26} For example, Chaurasia et al¹¹ used a convolutional neural network to

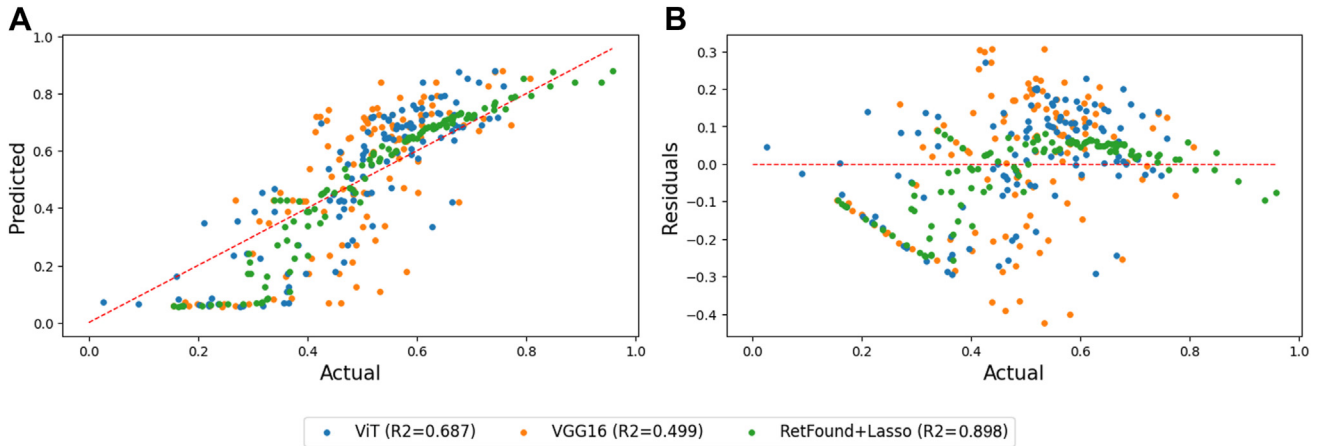


Figure 4. Regression analysis for cup-to-disc ratio (CDR). This figure shows scatter plot visualizations of regression results for models predicting CDR, including the best RETFound model and 2 baseline models (ViT and VGG16). **A**, The individual predicted model-predicted CDR against the actual CDR. **B**, The individual residuals for each sample. ViT = Vision Transformer.

Table 4. Clock Hours and Clock Quadrant Model Results

Model	MAE	MSE	RMSE	R ²
Clock Hours				
RetFound + Ridge	13.329	391.961	19.798	0.572
RetFound + Lasso	13.241	386.247	19.653	0.582
RetFound + ElasticNet	13.222	386.076	19.649	0.583
RetFound + OLS	27.533	1438.664	37.930	-0.661
Pretrained ViT fine-tuned	15.519	501.333	22.390	0.444
Pretrained VGG fine-tuned	17.742	635.238	25.204	0.340
Clock quads				
RetFound + ridge	3.442	61.592	7.848	0.809
RetFound + lasso	3.498	61.039	7.813	0.811
RetFound + ElasticNet	3.494	61.742	7.858	0.808
RetFound + OLS	8.115	227.663	15.089	0.292
Pretrained ViT fine-tuned	4.365	86.313	9.290	0.732
Pretrained VGG fine-tuned	7.361	174.999	13.229	0.668

MAE = mean absolute error; MSE = mean squared error; OLS = ordinary least squares; RMSE = root mean squared error; ViT = Vision Transformer.

Bolded values correspond to the metrics from the strongest-performing regression model.

estimate CDR from 181 768 fundus images. The authors report an R^2 score of 0.71 on their validation set; we achieved R^2 of 0.89 for the same task.¹¹ Medeiros et al previously trained a deep learning convolutional neural network on 32 820 pairs of optic disc photographs and RNFL OCT scans to predict average RNFL thickness, achieving R^2 of 0.693; we achieved R^2 of 0.961. Importantly, our methodology achieves high performance despite training on a smaller dataset, highlighting the advantage of using a pretrained foundation model. Our study illustrates that by using a pretrained retinal image foundation model as a feature extractor, we can achieve performance that is comparable to or surpasses deep learning systems specifically trained for single tasks. This approach offers several critical advantages, especially in clinical settings. In addition to reducing the need for curation and labeling of large dedicated imaging corpora, the computational time and resources required are also reduced, as the model weights for RETFound are frozen. These advantages can lower the barrier to developing and adopting artificial intelligence systems in health care or any resource-constrained environment. Finally, by

leveraging the rich, pretrained representations of the foundation model, our approach can also minimize the risk of overfitting and potentially enhance generalizability.

In addition to predicting CDR and average RNFL thickness, we also trained models to predict RNFL thickness at the quadrant and clock-hour level. These were multi-output prediction models, and performance was not as high as for predicting overall CDR or average RNFL thickness. Predicting RNFL thickness at such a granular level may be an inherently more difficult task as previous literature has shown there is an increased variability and lower repeatability in clock-hour RNFL thickness compared to quadrant or global RNFL thickness measurements, making them less reliable for tracking glaucoma progression.^{27–29} Increased complexity or granularity of the prediction may make it challenging for the models to accurately capture and identify appropriate patterns within the image features, leading to lower R^2 values and higher measurement error. Additionally, the task demands precise identification of features corresponding to each clock hour, which may be beyond the capability of the regression models using features extracted from the entire image. It is notable that performance for predicting RNFL thickness of superior and inferior locations was better than for nasal and temporal locations. This variation could be related to variation in vessel location or due to the effects of optic nerve tilt and peripapillary atrophy, which have greater effects along the horizontal meridian.

A strength of our study is analysis of performance stratified by race/ethnicity, which is not often reported by AI validation studies. Our models demonstrated that model performance was best on non-Hispanic White and Asian subpopulations for all tasks. This may be due to the demographic distribution of the population as these 2 groups are most highly represented in our cohort. Differences in eye morphology between different demographic groups could also lead to disparities in model performance. In addition, there were relatively few patients of other demographic subgroups (e.g., Black patients) within our cohort; model performance in these subgroups was not as strong but it is possible that results may be different if evaluated on larger groups of patients. It may also be possible that features extracted from RETFound also contain biases. Over 90% of the fundus images used to train the RETFound model are from Moorfields Eye Hospital; although exact demographics of this set have not been released, it may be likely that the

Table 7. R^2 Values Stratified by Race/Ethnicity

	# of Samples	RNFL Thickness	CDR	Clock Hours	Clock Quadrants
Hispanic	24	0.915	0.842	0.521	0.721
Non-Hispanic White	52	0.982	0.918	0.592	0.872
Non-Hispanic Asian	30	0.933	0.862	0.562	0.801
Non-Hispanic Black or African American	8	0.888	0.851	0.511	0.689
Other	36	0.734	0.721	0.532	0.665

CDR = cup-to-disc ratio; RNFL = retinal nerve fiber layer.

demographics reflect those of the UK and would have a minority of Black and Hispanic patients. Our regression models could also have introduced bias as using fine-tuning or training models on extracted features from a pretrained model has been known to lead to unfair outcomes.³⁰ Additional work is needed to further analyze the fairness of foundation models on downstream tasks and investigate bias mitigation techniques to predict outcomes fairly.

Several limitations remain in this study. Our validation relied upon data from a single academic center. Although models using RETFound performed very well on our clinical data, results may differ at other sites. Furthermore, the pairs of fundus images and RNFL OCT scans used for this study come from patients who are being referred for some ophthalmic concern; thus, patients with pathology or suspected pathology are overrepresented compared to normal patients. Suitability of such models for the purpose of screening asymptomatic patients requires further evaluation. A key limitation of this study is that the majority of the fundus images were captured using the same machine, which may limit the generalizability of the model to images from other devices as different OCT machines have different segmentations affecting CDR measurements. This could lead to biased predictions or reduced accuracy when applied

to fundus images obtained from different imaging systems. Further analysis is required to validate the model on fundus images generated on a wide variety of machines. Another limitation is that although fundus images and RNFL parameters were matched within a 6-month timeframe, this approach does not account for potential changes in retinal conditions or variations in RNFL measurements that might occur within that period. As a result, discrepancies between the OCT and RNFL measurements could impact the accuracy of the predictive model.

In conclusion, we have demonstrated that using foundation models like RETFound as a feature extractor can effectively predict various downstream regression tasks, performing comparably to deep learning systems trained on extensive data for a single task. Our proposed method enables clinicians to deploy these powerful models with minimal technical expertise, less data, and in a fraction of the time required to develop deep learning systems from scratch. We also explore how performance can significantly vary across different subpopulations and further work is needed to employ bias mitigation techniques in order to develop fair systems. Additionally, work is needed to explore the explainability of the foundation model on downstream tasks to build clinical trust.

Footnotes and Disclosures

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No animal subjects were used in this study.

Author Contributions:

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Data collection: Chen, Ravindranath, Wang

Analysis and interpretation: Chen, Ravindranath, Chang, Keane, Wang

Obtained Funding: Wang, Keane

Overall responsibility: Chen, Ravindranath, Chang, Zhou, Keane, Wang

Abbreviations and Acronyms:

CDR = cup-to-disc ratio; **MAE** = mean absolute error; **MSE** = mean squared error; **RMSE** = root mean squared error; **SD** = standard deviation; **ViT** = Vision Transformer.

Keywords:

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References

- Stein JD, Khawaja AP, Weizer JS. Glaucoma in adults—screening, diagnosis, and management: a review. *JAMA*. 2021;325:164–174.
- Chauhan BC, Malik R, Shuba LM, et al. Rates of glaucomatous visual field change in a large clinical population. *Invest Ophthalmol Vis Sci*. 2014;55:4135–4143.
- Wagner IV, Stewart MW, Dorairaj SK. Updates on the diagnosis and management of glaucoma. *Mayo Clin Proc Innov Qual Outcomes*. 2022;6:618–635.
- Law SK, Kornmann HL, Nilforushan N, et al. Evaluation of the "IS" rule to differentiate glaucomatous eyes from normal. *J Glaucoma*. 2016;25:27–32.
- Geevarghese A, Wollstein G, Ishikawa H, Schuman JS. Optical coherence tomography and glaucoma. *Annu Rev Vis Sci*. 2021;7:693–726.
- Sathyan P, Shilpa S, Anitha A. Optical coherence tomography in glaucoma. *J Curr Glaucoma Pract*. 2012;6:1–5.
- Haja SA, Mahadevappa V. Advancing glaucoma detection with convolutional neural networks: a paradigm shift in ophthalmology. *Rom J Ophthalmol*. 2023;67:222–237.
- Fan R, Alipour K, Bowd C, et al. Detecting glaucoma from fundus photographs using deep learning without convolutions: transformer for improved generalization. *Ophthalmol Sci*. 2023;3:100233.
- Akter N, Fletcher J, Perry S, et al. Glaucoma diagnosis using multi-feature analysis and a deep learning technique. *Sci Rep*. 2022;12:8064.
- Wang SY, Ravindranath R, Stein JD, et al. Prediction models for glaucoma in a multicenter electronic health records consortium: the sight outcomes Research collaborative. *Ophthalmol Sci*. 2024;4:100445.
- Chaurasia AK, Greatbatch CJ, Han X, et al. Highly accurate and precise automated cup-to-disc ratio quantification for glaucoma screening. *Ophthalmol Sci*. 2024;4:100540.
- Gao XR, Wu F, Yuhas PT, et al. Automated vertical cup-to-disc ratio determination from fundus images for glaucoma detection. *Sci Rep*. 2024;14:4494.
- Park K, Kim J, Lee J. Automatic optic nerve head localization and cup-to-disc ratio detection using state-of-the-art deep-learning architectures. *Sci Rep*. 2020;10:5025.
- Razaghi G, Hedayati E, Hejazi M, et al. Measurement of retinal nerve fiber layer thickness with a deep learning algorithm in ischemic optic neuropathy and optic neuritis. *Sci Rep*. 2022;12:17109.
- Tian Y, Shi M, Luo Y, et al. Predicting personalized near-normal retinal nerve fiber layer thickness for glaucoma patients with deep learning. *Invest Ophthalmol Vis Sci*. 2023;64:1128.
- Yang H, Ahn Y, Askaruly S, et al. Deep learning-based glaucoma screening using regional RNFL thickness in fundus photography. *Diagnostics (Basel)*. 2022;12:2894.
- Medeiros FA, Jammal AA, Thompson AC. From machine to machine: an OCT-trained deep learning algorithm for objective quantification of glaucomatous damage in fundus photographs. *Ophthalmology*. 2019;126:513–521.
- Krishnan R, Rajpurkar P, Topol EJ. Self-supervised learning in medicine and healthcare. *Nat Biomed Eng*. 2022;6:1346–1352.
- Moor M, Banerjee O, Abad ZSH, et al. Foundation models for generalist medical artificial intelligence. *Nature*. 2023;616:259–265.
- Zhou Y, Chia MA, Wagner SK, et al. A foundation model for generalizable disease detection from retinal images. *Nature*. 2023;622:156–163.
- Jeong Y, Hong YJ, Han JH. Review of machine learning applications using retinal fundus images. *Diagnostics (Basel)*. 2022;12:134.
- Kolesnikov A, Dosovitskiy A, Weissenborn D, et al. An image is worth 16x16 words: transformers for image recognition at scale. Presented at: Proceedings of the 9th International Conference on Learning Representations (ICLR) *ArXiv*. 2021. <https://doi.org/10.48550/arXiv.2010.11929>.
- Liu S, Deng W. *Very deep convolutional neural network based image classification using small training sample size*. 2015 3rd IAPR Asian Conference on Pattern Recognition (ACPR). Malaysia: Kuala Lumpur; 2015:730–734.
- Christopher M, Hoseini P, Walker E, et al. A deep learning approach to improve retinal structural predictions and aid glaucoma neuroprotective clinical trial design. *Ophthalmol Glaucoma*. 2023;6:147–159.
- Larrosa JM, Polo V, Ferreras A, et al. Neural network analysis of different segmentation strategies of nerve fiber layer assessment for glaucoma diagnosis. *J Glaucoma*. 2015;24:672–678.
- Mariottoni EB, Jammal AA, Urata CN, et al. Quantification of retinal nerve fibre layer thickness on optical coherence tomography with a deep learning segmentation-free approach. *Sci Rep*. 2020;10:402.
- Ghasia FF, El-Dairi M, Freedman SF, et al. Reproducibility of spectral domain optical coherence tomography measurements in adult and pediatric glaucoma. *J Glaucoma*. 2013;24:55–63.
- Mwanza JC, Chang RT, Budenz DL, Durbin MK. Reproducibility of peripapillary retinal nerve fiber layer thickness and optic nerve head parameters measured with cirrus HD-OCT in glaucomatous eyes. *Invest Ophthalmol Vis Sci*. 2010;51:5724–5730.
- Asrani S, Essaid L, Alder BD, Santiago-Turla C. Artifacts in spectral-domain optical coherence tomography measurements in glaucoma. *JAMA Ophthalmol*. 2014;132:396–402.
- Zhang Y, Zhou F. Bias mitigation in fine-tuning pre-trained models for enhanced fairness and efficiency. *arXiv*. 2024. <https://doi.org/10.48550/arXiv.2403.00625>.